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Sakthi Kanagam
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Objective:

AI/ML Engineer with 6 months of practical experience in developing backend systems and intelligent features using Generative AI, Large Language Models and Python-based frameworks. Skilled in LLM fine-tuning, prompt engineering, and building multi-agent systems using tools like LangChain, LangGraph, and CrewAI. Proficient in deploying machine learning and deep learning models to automate decisions and enhance business intelligence, especially in financial and data-driven domains.

Technical Skills:

Programming: Python (AI, ML, DL), SQL, HTML, CSS, JavaScript

GenAI & LLM Ecosystem: LangChain, LangGraph, CrewAI, OpenAI API, Hugging Face Transformers

Prompt Engineering: Zero-shot, Few-shot, Chain-of-Thought, RAG pipelines, Agent orchestration

Machine Learning: scikit-learn, XGBoost, pandas, NumPy

Deep Learning & CV: TensorFlow, PyTorch, OpenCV, CNNs, RNNs

Backend & APIs: Django REST Framework, FastAPI

Databases: MySQL, Firebase

Development Tools: Git, Jupyter Notebook, VS Code, Postman

Work Experience:

Associating with **SPIRO PRIME TECH SERVICES**

Project: Smart Help – AI-Driven HR Management System using RAG and Multi-Agent LLMs

Technologies: Python 3.10, Django, FastAPI, LangChain, LangGraph, CrewAI, SQLite, OpenAI API, Hugging Face Transformers, Scikit-learn, Pandas, OpenCV

Tools: VS Code, Git, Jupyter Notebook, Postman, Docker.

Period: 10/2024 to 07/2025.

Description: An intelligent HR management system powered by LLMs, retrieval-augmented generation and AI agents to automate HR-related tasks such as onboarding, meeting scheduling, and policy-based query resolution. Built to simulate an HR assistant capable of interpreting employee requests and generating actionable responses in real-time. The system replaces traditional HR support for onboarding interns, managing schedules, and planning team events by using orchestrated multi-agent frameworks and internally stored knowledge.

Responsibilities:

✓ Designed and developed a multi-agent framework using LangChain, CrewAI, and LangGraph to automate HR workflows like onboarding, scheduling meetings, and planning welcome events.

✓ Created custom AI agents such as Query Analyzer, Scheduler, Event Planner, and Communication Agent to break down and execute complex HR tasks.

✓ Implemented RAG pipelines by integrating LangChain with internal HR documents and FAQs stored in a vector database for context-aware question answering.

✓ Fine-tuned LLMs using Hugging Face Transformers and created prompt templates tailored for HR

use cases.

✓ Developed backend services using FastAPI and Django REST Framework to expose model outputs via secure internal APIs.

✓ Used OpenCV for document preprocessing during employee verification and ID parsing workflows.

✓ Managed end-to-end data processing, including cleaning, annotation, and categorization of HR use cases into structured intents.

✓ Deployed and tested the system in a sandboxed internal environment with restricted internet access to simulate real enterprise use.

Own Projects:

Project: Real-Time Ambulance Detection Using Camera and Audio Feeds (Final Year)

Technologies:(React JS, HTML5, CSS3, Bootstrap, Python 3.8, OpenCV, TensorFlow, Librosa, REST API, YOLOv5, NumPy, SQLite)

Tools:(Jupyter Notebook, VS Code, Git, Docker, Postman, Google Colab, Agile, GitHub, VLC)

Description: Ambulance Tracking is an intelligent system designed to detect the presence of ambulances in real-time using both live camera feeds and audio streams. The system was created to alert traffic signals or emergency units to prioritize ambulance movement during congestion. It works offline and supports integration with local smart traffic controllers, develop robust, scalable, Responsive and high-performance web applications using Django.

Responsibilities:

✓ Developed and maintained high-quality software modules using Python and React.

✓ Wrote clean, modular, and maintainable code using HTML, CSS, and JavaScript for frontend development.

✓ Collected and labeled over 300+ time audio recordings of ambulance sirens and 10,000+ traffic images for model training

✓ Processed video feeds using OpenCV and trained YOLOv5 for vehicle detection. Extracted MFCC and spectral features using Librosa, and built a CNN model to classify siren audio.

Educational Details:

Bachelor of Technology in Information Technology

St. Joseph College of Engineering, Sriperumbudur, Chennai

Duration: 2020 – 2024